EX.No.-2.2 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

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AIM:

To create a 3D model from the given isometric drawing using fusion 360 tool.

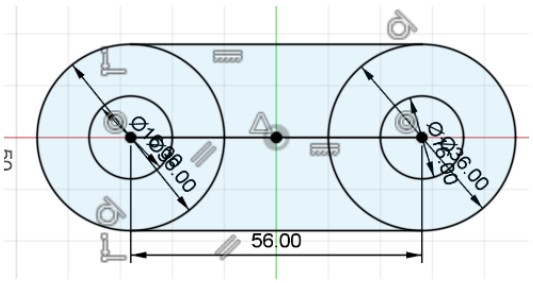
REQUIREMENTS:

Autodesk fusion 360

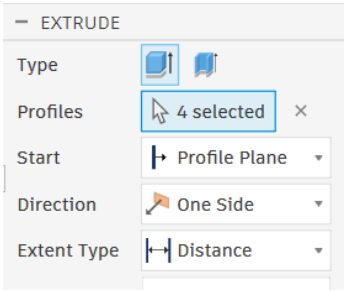
1. Windows 10.
2. 16 GB of RAM (integrated graphics recommend 1 GB or more).
3. 2.5 Mbps or faster download; 500 Kbps or faster upload.
4. 2.5 Mbps or faster download; 500 Kbps or faster upload.

PROCEDURE:

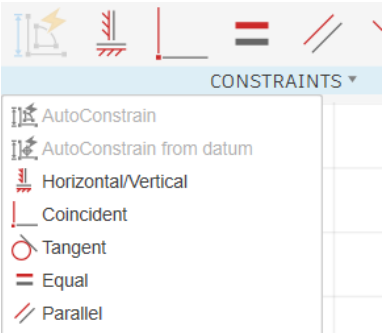
1. Open Autodesk Fusion 360 and click on New Design. Select Create Sketch and choose the Top Plane. Draw a horizontal line of 56 mm and draw two concentric circles at both endpoints of the line to form the base flange. The inner circles should have a diameter of 16 mm, and the outer circles should have a radius of 18 mm. Now draw two horizontal lines above and below and join them with the two circles by using a tangent tool.Trim the rest of the lines.



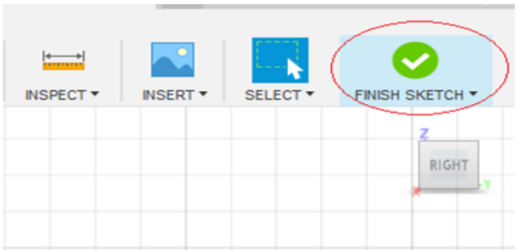

2. After completing the sketch, click on Finish Sketch and select Extrude. Extrude the required profile to a distance of 8 mm. Now, select the top face of the object and click on Create Sketch. Using the Two-Point Rectangle tool, draw a rectangle on the previously created object with a horizontal dimension of 40 mm and a vertical dimension of 10 mm.




3. Click on Finish Sketch and extrude the rectangular profile 32 mm upwards. Again, select the top face and click on Create Sketch. Draw a vertical line of 32 mm from the 40 mm horizontal line. Using the endpoint of the vertical line as the centre, draw two concentric circles with an inner diameter of 20 mm and an outer diameter of 30 mm.
4. Draw vertical lines from both ends of the 40 mm horizontal line and use the Tangent Constraint tool to connect the lines tangentially to the outer circle of diameter 30 mm. Then, use the Trim tool to remove the unwanted portions of the sketch.

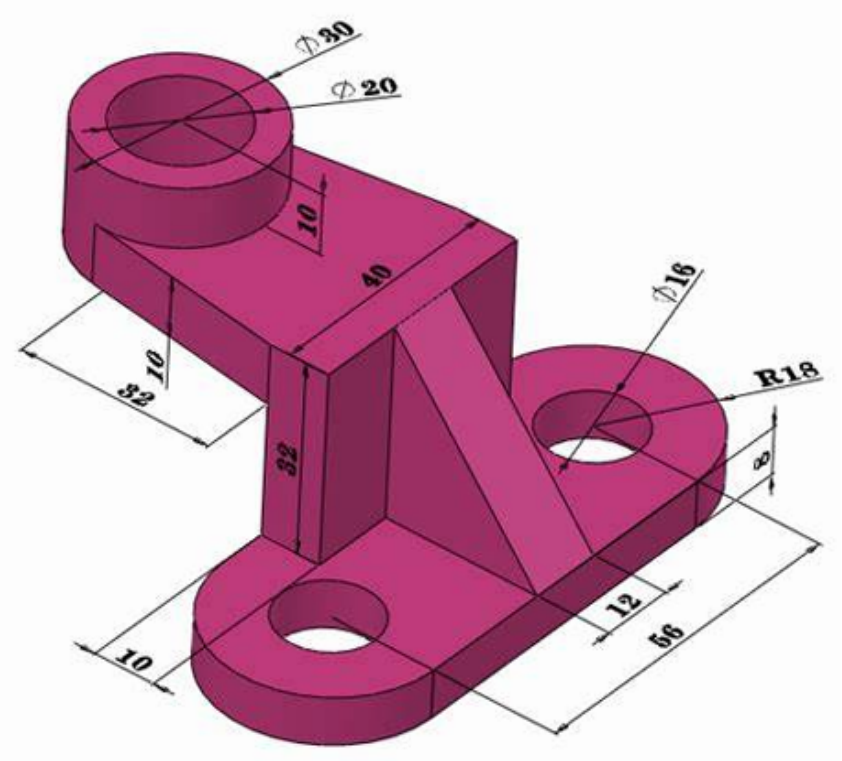



5. Click on Finish Sketch and select Extrude. Extrude the required profile to a distance of 10 mm downwards. After extruding, select the top face of the extruded part and click on Create Sketch. Draw a circle of diameter 30 mm at the required position. Click on Finish Sketch and extrude the circular profile to a distance of 10 mm upwards, keeping the inner circular opening of diameter 20 mm.
6. Now, select Create Sketch and choose the required face of the base flange. Using the Two-Point Rectangle tool, draw a rectangle with a horizontal dimension of 12 mm and a vertical dimension of 26 mm. Click on Finish Sketch and extrude the rectangular profile to a distance of 32mm upwards.

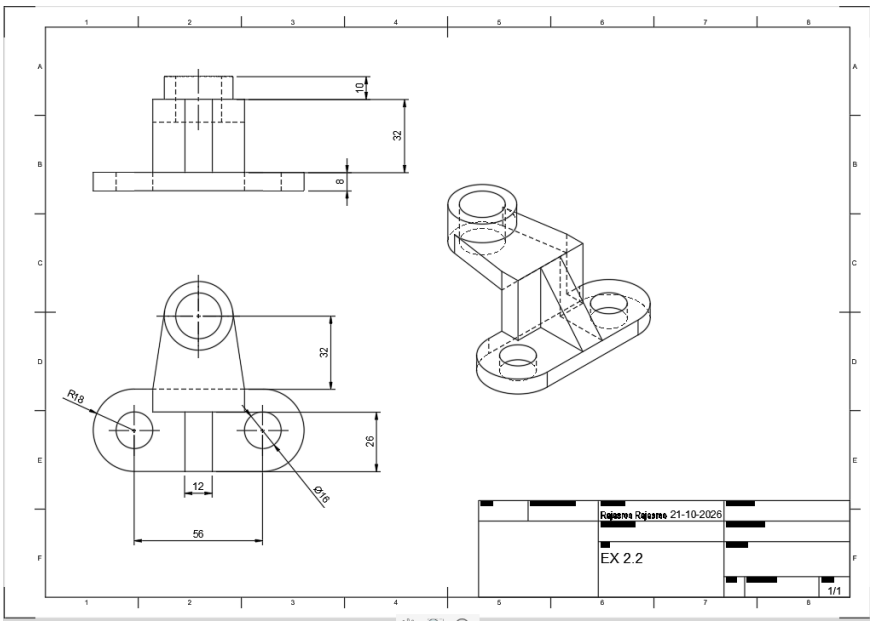



7. After completing the extrusion, go to the Modify menu and select the Chamfer tool. Choose the required chamfer option, select the appropriate edges, and apply the chamfer with the specified dimensions. Click OK to complete the operation.
8. Finally, check all the dimensions and features of the completed 3D model and save the design.Finally select the drawing menu and click from design and start giving the dimensions.

   INPUT:




   OUTPUT:




RESULT:

Thus, the 3D model is created from given isometric drawing.